

Notes: Hahn, Todd, and Van der Klaauw (Econometrica, 2001)
Identification and Estimation of Treatment Effects with a
Regression-Discontinuity Design

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1 Big picture

1.1 Main question

The paper asks what, exactly, a regression-discontinuity design identifies and how the resulting treatment effect should be estimated with minimal functional-form assumptions. The setting is a treatment that changes discontinuously at a known cutoff in a running variable. The object of interest is not necessarily an average treatment effect for the full population. Instead, the RD design identifies a treatment effect local to the cutoff.

This distinction is the core message. A discontinuity in treatment assignment gives quasi-experimental variation only for observations arbitrarily close to the threshold. Observations far from the cutoff may be informative for precision or functional-form extrapolation, but the nonparametric identification argument itself is local.

1.2 Why this paper matters

Before this paper, many RD applications imposed stronger assumptions than necessary, often using parametric regression models or treating the assignment variable like an ordinary instrument. This note shows that the key identifying restriction is weaker and more local: the conditional expectation of the potential outcome without treatment must be continuous at the cutoff. The discontinuity in treatment assignment is then used to isolate the treatment effect.

The paper also clarifies the difference between sharp and fuzzy RD. In a sharp RD, treatment status jumps deterministically from 0 to 1 at the cutoff, so the treatment effect is the jump in the outcome regression function. In a fuzzy RD, treatment probability jumps but not necessarily from 0 to 1, so the treatment effect is the ratio of the jump in outcomes to the jump in treatment probability.

1.3 Main lesson

The paper’s main lesson can be summarized as:

$$\text{RD effect at cutoff} = \frac{\text{jump in } E[Y \mid Z = z] \text{ at } z_0}{\text{jump in } E[X \mid Z = z] \text{ at } z_0}.$$

For sharp RD, the denominator equals one. For fuzzy RD, the denominator is the first-stage discontinuity. The denominator matters because the outcome jump must be scaled by the amount of treatment induced at the cutoff.

2 Incremental contribution of the paper

2.1 Formal nonparametric identification for RD

The first contribution is to state RD identification in a clean nonparametric potential-outcome framework. The paper shows that the treatment effect is identified by comparing one-sided limits at the cutoff, not by imposing a global linear model or assuming treatment assignment is independent of unobservables everywhere.

2.2 Sharp RD and fuzzy RD in one framework

The paper treats sharp RD as a special case of fuzzy RD. In the sharp case, the jump in treatment probability is exactly one. In the fuzzy case, the treatment probability changes discontinuously but incompletely. This unified framework made it easier to connect RD to instrumental-variables logic while still emphasizing that the RD argument is local.

2.3 Heterogeneous treatment effects and the cutoff-specific estimand

The paper moves beyond constant treatment effects. With treatment-effect heterogeneity, the identified parameter is the average treatment effect at the cutoff under conditional independence. Without conditional independence, but with monotonic treatment response to the cutoff rule, the identified parameter is a local average treatment effect for individuals whose treatment status changes at the threshold.

2.4 Why continuity is doing the identifying work

A key conceptual contribution is the demonstration that an ordinary zero conditional mean restriction is not enough. The running variable can be correlated with unobservables. What matters is that the conditional expectation of the untreated potential outcome does not jump at the same cutoff. The discontinuity in treatment probability is useful only because the untreated outcome path is assumed to be smooth through the threshold.

2.5 Estimation: local Wald and local linear RD

The paper connects RD estimation to a local Wald estimator and explains why inference is different from standard IV inference. It then argues for local linear regression because ordinary kernel regression performs poorly at boundaries. In RD, every one-sided limit is a boundary estimate, so boundary bias is central rather than incidental.

3 Model and notation

3.1 Potential outcomes

Let Y_{1i} be the outcome for individual i if treated and Y_{0i} the outcome if untreated. Let $X_i \in \{0, 1\}$ denote treatment status. The observed outcome is

$$Y_i = \alpha_i + X_i\beta_i,$$

where $\alpha_i = Y_{0i}$ and $\beta_i = Y_{1i} - Y_{0i}$.

The evaluation problem is that we never observe both Y_{1i} and Y_{0i} for the same person at the same time. RD solves this problem locally by exploiting a discontinuity in the probability of treatment.

3.2 Running variable and cutoff

Let Z_i be the running variable, and let z_0 be the known cutoff. The treatment assignment mechanism satisfies

$$x^+ = \lim_{z \downarrow z_0} E[X_i | Z_i = z], \quad x^- = \lim_{z \uparrow z_0} E[X_i | Z_i = z],$$

with $x^+ \neq x^-$. The notation here means the conditional treatment probability has a jump at z_0 .

3.3 Sharp versus fuzzy designs

In a **sharp RD**, treatment is a deterministic function of the running variable: for example, everyone above the cutoff is treated and everyone below is not. Then $x^+ = 1$ and $x^- = 0$.

In a **fuzzy RD**, crossing the cutoff changes treatment probability but does not perfectly determine treatment. Some people below the cutoff may receive treatment and some above may not. The discontinuity is still useful as long as $x^+ \neq x^-$.

4 Identification results

4.1 Constant treatment effects

If treatment effects are constant, $\beta_i = \beta$, and if $E[\alpha_i | Z_i = z]$ is continuous at z_0 , then

$$\beta = \frac{y^+ - y^-}{x^+ - x^-},$$

where

$$y^+ = \lim_{z \downarrow z_0} E[Y_i | Z_i = z], \quad y^- = \lim_{z \uparrow z_0} E[Y_i | Z_i = z].$$

The numerator is the jump in outcomes at the cutoff. The denominator is the jump in treatment probability. Continuity of the untreated outcome makes the non-treatment part cancel out when taking the difference in one-sided limits.

4.2 Sharp RD as a special case

In a sharp design, $x^+ = 1$ and $x^- = 0$, so the estimand simplifies to

$$\beta = y^+ - y^-.$$

The sharp RD treatment effect is just the discontinuity in the conditional expectation of the outcome at the threshold.

4.3 Variable treatment effects

When treatment effects vary across individuals, the constant-effect estimand is no longer sufficient. If $E[\beta_i | Z_i = z]$ is continuous at z_0 and treatment status is conditionally independent of individual treatment effects near the cutoff, the same ratio identifies

$$E[\beta_i | Z_i = z_0] = \frac{y^+ - y^-}{x^+ - x^-}.$$

This is the average treatment effect for individuals located at the cutoff.

4.4 LATE interpretation without conditional independence

The paper then relaxes the conditional independence assumption by allowing people or groups to have different deterministic assignment functions. Under monotonicity - the cutoff weakly increases treatment status for everyone near the threshold - the RD Wald ratio identifies a local average treatment effect:

$$\lim_{\varepsilon \rightarrow 0^+} E[\beta_i \mid x_i(z_0 + \varepsilon) - x_i(z_0 - \varepsilon) = 1] = \frac{y^+ - y^-}{x^+ - x^-}.$$

The conditioning event isolates the subgroup whose treatment status changes discontinuously at the cutoff. These are the RD analog of compliers.

4.5 What is not identified

Without further assumptions, RD does not identify treatment effects away from the cutoff. The design identifies a local parameter because the identifying comparison is between units infinitesimally close to z_0 . Extrapolating beyond the cutoff requires extra assumptions, such as a parametric functional form or stable treatment effects across values of Z .

5 Estimation

5.1 The basic estimator

The identified parameter is estimated by replacing the one-sided limits with nonparametric estimates:

$$\hat{\beta}_{RD} = \frac{\hat{y}^+ - \hat{y}^-}{\hat{x}^+ - \hat{x}^-}.$$

The main estimation problem is therefore estimating four limits at the boundary: two outcome limits and two treatment-probability limits.

5.2 Local Wald interpretation

Using one-sided uniform kernel estimators, the RD estimator is numerically equivalent to an IV/Wald estimator estimated on observations in a small window around the cutoff, with an indicator for being above the cutoff used as an instrument.

This equivalence is useful, but the interpretation differs from ordinary IV. RD does not require the running variable to be uncorrelated with the error term globally. Instead, RD requires that the relevant conditional expectations be continuous at the cutoff. The instrument is valid locally because crossing the threshold creates a discontinuous change in treatment while the other determinants of outcomes are assumed to evolve smoothly.

5.3 Boundary bias and local linear regression

The paper emphasizes that standard kernel regression has poor finite-sample performance at boundaries. In a standard interior nonparametric regression problem, bias is often of order $O(h^2)$. At a

boundary, the bias can be of order $O(h)$. RD estimation is entirely about boundary points because the left and right limits are estimated at the cutoff.

The authors therefore recommend local linear regression. For the right-hand outcome limit, local linear regression solves

$$(\hat{a}, \hat{b}) = \arg \min_{a,b} \sum_{i=1}^n (Y_i - a - b(Z_i - z_0))^2 K\left(\frac{Z_i - z_0}{h}\right) 1(Z_i > z_0),$$

and estimates y^+ by $\hat{a}$. Analogous regressions estimate y^- , x^+ , and x^- . Local linear regression reduces boundary bias and makes RD estimation more reliable.

6 Practical interpretation

6.1 What the cutoff comparison means

The RD comparison is valid when units just above and just below the cutoff are comparable except for the discontinuous change in treatment probability. This does not require the running variable to be randomly assigned everywhere. It only requires that potential outcomes do not themselves jump at the cutoff.

6.2 What should be checked in applications

The paper implies the following empirical checklist:

1. Verify that treatment probability changes discontinuously at the cutoff.
2. Estimate the jump in the outcome using observations close to the threshold.
3. Use local polynomial methods, especially local linear regression, rather than relying on global functional forms.
4. Interpret the estimate as local to the cutoff.
5. In fuzzy RD, scale the outcome jump by the treatment-probability jump.
6. Be cautious about treatment-effect heterogeneity: the estimand may be a cutoff-specific ATE or a LATE for threshold compliers, depending on assumptions.

6.3 Connection to modern RD practice

Modern RD practice develops bandwidth selection, robust bias correction, placebo tests, density tests, and covariate balance checks. This paper is foundational because it clarifies the target estimand and the boundary-estimation problem that these later methods refine.

7 Interpretation of theorem and equation

The paper has no formal data tables or plotted empirical figures. Following the same visual-note standard, the extracted images below reproduce the main theorem and equation blocks from the paper. They are placed two per row where possible, and the final single extract uses `width=0.6\textwidth`.

7.1 Extracts from the paper

2. REGRESSION-DISCONTINUITY DESIGN AND SOURCES OF IDENTIFICATION

The goal of an evaluation is to determine the effect that some binary treatment variable x_i has on an outcome y_i . The evaluation problem arises because persons either receive or do not receive treatment and no individual is observed in both states at the same time. Let y_{1i} denote the outcome with treatment and y_{0i} that in the absence of treatment, and let $x_i = 1$ if treatment is received and $x_i = 0$ otherwise. The model for the observed outcome can be written as $y_i = \alpha_i + x_i \cdot \beta_i$, where $\alpha_i \equiv y_{0i}$, $\beta_i \equiv y_{1i} - y_{0i}$.

There are two main types of discontinuity designs considered in the literature—the sharp design and the so-called fuzzy design.² With a sharp design, treatment x_i is known to depend in a deterministic way on some observable variable z_i , $x_i = f(z_i)$, where z_i takes on a continuum of values and the point z_0 , where the function $f(z)$ is discontinuous, is assumed to be known. With a fuzzy design, x_i is a random variable given z_i , but the conditional probability $f(z) \equiv E[x_i | z_i = z] = \Pr[x_i = 1 | z_i = z]$ is known to be discontinuous at z_0 . The fuzzy design differs from the sharp design in that the treatment assignment is not a deterministic function of z_i ; there are additional variables unobserved by the econometrician that determine assignment to treatment. The common feature it shares with the sharp design is that the probability of receiving treatment, $\Pr[x_i = 1 | z_i]$, viewed as a function of z_i , is discontinuous at z_0 :

ASSUMPTION (RD): (i) The limits $x^+ \equiv \lim_{z \rightarrow z_0^+} E[x_i | z_i = z]$ and $x^- \equiv \lim_{z \rightarrow z_0^-} E[x_i | z_i = z]$ exist.³ (ii) $x^+ \neq x^-$.

Extract 1. RD setup and Assumption RD.

2.1. Constant Treatment Effects

Suppose that the treatment effect β is constant across different individuals. Let $e > 0$ denote an arbitrary small number. Suppose that we have a reason to believe that in the absence of treatment, persons close to the threshold z_0 are similar. We would then expect $E[\alpha_i | z_i = z_0 + e] \approx E[\alpha_i | z_i = z_0 - e]$, which motivates the assumption:

ASSUMPTION (A1): $E[\alpha_i | z_i = z]$ is continuous in z at z_0 .

Below, we establish that β is nonparametrically identified solely under this continuity restriction:

THEOREM 1: Suppose that β_i is fixed at β . Further suppose that Assumptions (RD) and (A1) hold. We then have

$$(1) \quad \beta = \frac{y^+ - y^-}{x^+ - x^-},$$

where $y^+ \equiv \lim_{z \rightarrow z_0^+} E[y_i | z_i = z]$ and $y^- \equiv \lim_{z \rightarrow z_0^-} E[y_i | z_i = z]$.

² See Trochim (1984).

³ Throughout this paper, we also assume that the density of z_i is positive in the neighborhood containing z_0 .

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Extract 2. Constant-effect identification.

Interpretation. Extract 1 defines the core RD environment. The important point is that sharp and fuzzy RD share the same identifying feature: a discontinuity in treatment probability at a known cutoff. Extract 2 shows the simplest identification result. If untreated outcomes are continuous and treatment effects are constant, the treatment effect is a Wald ratio of outcome and treatment jumps. In a sharp RD, the denominator is one.

NOW we consider the question of identification when treatment effects are heterogeneous. To generalize the identification strategy in the constant treatment effect case, we make the following assumption:

ASSUMPTION (A2): $E[\beta_i | z_i = z]$, regarded as a function of z , is continuous at z_0 .

We establish that the average treatment effect at z_0 , $E[\beta_i | z_i = z_0]$, is nonparametrically identified under the functional form restriction and a weak form of conditional independence:

THEOREM 2: Suppose that x_i is independent of β_i conditional on z_i near z_0 . Further suppose that Assumptions (RD), (A1), and (A2) hold. We then have

$$(3) \quad E[\beta_i | z_i = z_0] = \frac{y^+ - y^-}{x^+ - x^-}.$$

Extract 3. Variable-effect cutoff ATE.

ASSUMPTION (A3): (i) $(\beta_i, x_i(z))$ is jointly independent of z_i near z_0 . (ii) There exists $e > 0$ such that $x_i(z_0 + e) \geq x_i(z_0 - e)$ for all $0 < e < e$.

THEOREM 3: Suppose that Assumptions (RD), (A1), and (A3) hold. We then have

$$(5) \quad \lim_{e \rightarrow 0^+} E[\beta_i | x_i(z_0 + e) - x_i(z_0 - e) = 1] = \frac{y^+ - y^-}{x^+ - x^-}.$$

PROOF: Invoking the reasoning in Imbens and Angrist (1994), we obtain

$$E[x_i \cdot \beta_i | z_i = z_0 + e] - E[x_i \cdot \beta_i | z_i = z_0 - e] \\ = E[\beta_i | x_i(z_0 + e) - x_i(z_0 - e) = 1]$$

Extract 4. LATE interpretation.

Interpretation. Extract 3 extends the result to heterogeneous treatment effects. The same Wald ratio identifies the average treatment effect at the cutoff if treatment status is conditionally independent of individual gains near the threshold. Extract 4 provides the RD version of the LATE result: under monotonicity, the estimand is the average effect for individuals whose treatment changes at the cutoff. This is the fuzzy RD analog of the compliers in instrumental variables.

3. ESTIMATION

For both the sharp design and fuzzy design, the ratio

$$(6) \quad \frac{y^+ - y^-}{x^+ - x^-}$$

identifies the treatment effect at $z = z_0$. Thus, given consistent estimators $\hat{y}^+$, $\hat{y}^-$, $\hat{x}^+$, and $\hat{x}^-$ of the four one-sided limits in (6), the treatment effect can be consistently estimated by

$$\frac{\hat{y}^+ - \hat{y}^-}{\hat{x}^+ - \hat{x}^-}.$$

In principle, we can use any nonparametric estimator to estimate the limits. We first consider one-sided kernel estimation and observe that under certain conditions an estimate based on kernel regression will be numerically equivalent to a standard Wald estimator. We then argue that such an estimator may have a poor finite sample property due to the boundary problem and propose to avoid the boundary problem by using local linear nonparametric regression (LLR) methods.

Consider the special case where we use kernel regression estimators based on one-sided uniform kernels. For the uniform kernel, it is not difficult to show that

$$\begin{aligned} \hat{y}^+ &= \frac{\sum_{i \in \mathcal{S}} y_i \cdot w_i}{\sum_{i \in \mathcal{S}} w_i}, & \hat{y}^- &= \frac{\sum_{i \in \mathcal{S}} y_i \cdot (1 - w_i)}{\sum_{i \in \mathcal{S}} (1 - w_i)}, \\ \hat{x}^+ &= \frac{\sum_{i \in \mathcal{S}} x_i \cdot w_i}{\sum_{i \in \mathcal{S}} w_i}, & \hat{x}^- &= \frac{\sum_{i \in \mathcal{S}} x_i \cdot (1 - w_i)}{\sum_{i \in \mathcal{S}} (1 - w_i)}, \end{aligned}$$

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Extract 5. Why continuity is necessary.

Extract 6. Estimation by one-sided limits.

Interpretation. Extract 5 is conceptually important. It shows that the usual regression restriction $E[v_i | Z_i] = 0$ does not, by itself, identify the treatment effect. RD needs the specific continuity condition at the cutoff. Extract 6 turns the identification formula into an estimator and shows how one-sided kernel estimation becomes numerically equivalent to a local Wald estimator when a uniform kernel is used.

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where $\mathcal{S}$ denotes the subsample such that $z_0 - h < z_i < z_0 + h$, $w_i \equiv 1$ ($z_0 < z_i < z_0 + h$), and $h > 0$ denotes the bandwidth. The estimator is *numerically* equivalent to an IV estimator for the regression of y_i on x_i which uses w_i as an instrument, applied to the subsample $\mathcal{S}$. Denote this estimator by $\hat{\beta}_w$.

It is interesting to note that the regression discontinuity can ‘justify’ a Wald estimator even when the standard IV assumption is violated. To see this, put the model in more familiar econometric notation by writing $\alpha_i = E[\alpha_i] + v_i = \alpha + v_i$. Under the common treatment assumption, this yields a model

$$y_i = \alpha + x_i \cdot \beta + v_i.$$

Identification of β does not require that the error term v_i be uncorrelated with z_i . All that is required is continuity assumption (A1). As long as the researcher is willing to change h appropriately as a function of the sample size, $\hat{\beta}_w$ is consistent. Thus $\hat{\beta}_w$ is motivated by a different principle than is the usual Wald estimator, but for a particular choice of kernel and subsample they are numerically equivalent.

Although $\hat{\beta}_w$ is numerically equivalent to a local Wald estimator, inference based on $\hat{\beta}_w$ will be different from that based on a Wald estimator. $\hat{\beta}_w$ will be asymptotically biased, as are many other nonparametric-regression-based estimators, whereas the Wald estimator is asymptotically unbiased by assumption. The bias problem is exacerbated in the regression-discontinuity case due to the bad boundary behavior of the kernel regression estimator: at boundary points, the bias of the kernel regression estimator converges to zero at a slower rate than at interior points. Under conventional assumptions on the kernel function, the order of the bias of the standard kernel estimator is $O(h)$ at boundary points and $O(h^2)$ at interior points.⁶ For our problem, all the points of estimation are at boundaries, so the bias could be substantial in finite samples.⁷ It would be misleading to use the conventional confidence interval based on the asymptotic distribution of the (asymptotically unbiased) Wald estimator as the true coverage probability would be very different from the nominal coverage probability.

Because of the poor boundary performance of standard kernel estimators, we propose instead to estimate the limits by local linear regression (LLR), shown by Fan (1992) to have better boundary properties than the traditional kernel regression estimator. The local linear estimator for y^+ , for example, is given by $\hat{a}$, where

$$(\hat{a}, \hat{b}) \equiv \operatorname{argmin}_{a, b} \sum_{i=1}^n (y_i - a - b(z_i - z_0))^2 K\left(\frac{z_i - z_0}{h}\right) 1(z_i > z_0).$$

Extract 7. Wald equivalence and boundary bias.

Extract 8. Local linear estimator.

Interpretation. Extract 7 explains why the RD-Wald analogy should not be overread. The estimator may look like IV, but the logic is different because validity comes from local smoothness at the cutoff, not from global exclusion or zero-correlation assumptions. The same extract also explains why boundary bias is severe in RD. Extract 8 presents local linear regression as the solution: estimate one-sided intercepts with a local slope, then form the ratio of outcome and treatment discontinuities.

- (ii) The limits $m^+(z_0)$, $m^-(z_0)$, $m'^+(z_0)$, $m'^-(z_0)$, $m''^+(z_0)$, $m''^-(z_0)$, $p^+(z_0)$, $p^-(z_0)$, $p'^+(z_0)$, $p'^-(z_0)$, $p''^+(z_0)$, and $p''^-(z_0)$ exist and are finite.
- (iii) The density of z_i , $f(z)$, is continuous and bounded near z_0 . It is also bounded away from zero near z_0 .
- (iv) $K(\cdot)$ is continuous, symmetric, and nonnegative-valued with compact support.
- (v) $\sigma^2(z_i) = \text{var}(y_i | z_i)$ is uniformly bounded near z_0 . Similarly, $\eta(z_i) = \text{cov}(y_i, x_i | z_i)$ is uniformly bounded near z_0 . Furthermore, the limits $\sigma^{2+}(z_0)$, $\sigma^{2-}(z_0)$, $\eta^+(z_0)$, and $\eta^-(z_0)$ exist and are finite.
- (vi) $\lim_{z \rightarrow z_0^+} E[|y_i - m(z_i)|^3 | z_i = z]$ and $\lim_{z \rightarrow z_0^-} E[|y_i - m(z_i)|^3 | z_i = z]$ exist and are finite.
- (vii) The bandwidth sequence satisfies $h = h_n = \varrho \cdot n^{-1/5}$ for some ϱ . Then,

$$n^{\frac{2}{3}} \left(\frac{\hat{y}^+ - \hat{y}^-}{\hat{x}^+ - \hat{x}^-} - \frac{y^+ - y^-}{x^+ - x^-} \right) \rightarrow \mathcal{N}(\mu_f, \Omega_f),$$

where

$$\mu_f \equiv \frac{1}{x^+ - x^-} (\rho^+ m^{n+}(z_0) - \rho^- m^{n-}(z_0)) - \frac{y^+ - y^-}{(x^+ - x^-)^2} (\rho^+ p^{n+}(z_0) - \rho^- p^{n-}(z_0)),$$

and

$$\begin{aligned} \Omega_f \equiv & \frac{1}{(x^+ - x^-)^2} (\omega^+ \sigma^{2+}(z_0) + \omega^- \sigma^{2-}(z_0)) \\ & - 2 \frac{y^+ - y^-}{(x^+ - x^-)^3} (\omega^+ \eta^+(z_0) + \omega^- \eta^-(z_0)) \\ & + \frac{(y^+ - y^-)^2}{(x^+ - x^-)^4} (\omega^+ p^+(z_0)(1 - p^+(z_0)) + \omega^- p^-(z_0)(1 - p^-(z_0))), \end{aligned}$$

and where

$$\begin{aligned} \rho^+ \equiv & \frac{(\int_0^\infty u^2 K(u) du)^2 - (\int_0^\infty u^2 K(u) du)(\int_0^\infty u K(u) du)}{(\int_0^\infty u^2 K(u) du)(\int_0^\infty K(u) du) - (\int_0^\infty u K(u) du)^2} \frac{\varrho^2}{2}, \\ \omega^+ \equiv & \frac{\int_0^\infty ((\int_0^\infty s^2 K(s) ds) - (\int_0^\infty s K(s) ds) \cdot u)^2 K(u)^2 du}{f(z_0) \varrho \cdot [(\int_0^\infty u^2 K(u) du)(\int_0^\infty K(u) du) - (\int_0^\infty u K(u) du)^2]}. \end{aligned}$$

with ρ^- and ω^- similarly defined but now with the integral of integration over $(-\infty, 0)$.

PROOF: A derivation of the distribution of the estimator is available in Hahn, Todd, and Van der Klaauw (1999), or upon request from the authors.

Extract 9. Asymptotic distribution of the local-linear RD estimator.

Interpretation. Extract 9 gives the asymptotic distribution of the ratio estimator when the four limits are estimated by local linear regression. The expression is mathematically dense, but its role is straightforward: it justifies formal inference after estimating the two outcome limits and two treatment-probability limits. The theorem also makes clear that estimation error comes from both the numerator and denominator of the fuzzy RD ratio.

8 What later papers build on

8.1 Bandwidth and bias correction

This paper explains why local linear regression is attractive at the boundary but does not deliver the modern bandwidth and robust-bias-correction toolkit. Later RD work builds directly on this foundation by developing data-driven bandwidth selection, bias-corrected confidence intervals, and sensitivity analysis across bandwidth choices.

8.2 Validity diagnostics

The identification argument implies diagnostic tests but does not fully develop them. Later practice often checks whether predetermined covariates are continuous at the cutoff, whether the density of

the running variable is continuous, and whether results are robust to different polynomial orders and bandwidths. These diagnostics operationalize the smoothness assumptions emphasized here.

8.3 Local interpretation

Modern applications increasingly emphasize that RD estimates are local to the cutoff. This paper is an early formal statement of that point: without extra assumptions, the identified parameter is the treatment effect for the subgroup located at the threshold, not a population-wide effect.

9 Limitations and cautions

9.1 External validity

RD identifies treatment effects at the threshold. If treatment effects vary with the running variable, the cutoff effect may not apply to units far from the cutoff. This is not a flaw of the design but a feature of what the design credibly identifies.

9.2 Continuity is untestable in full

The key assumption is that potential outcomes would evolve smoothly through the cutoff absent treatment. Observed covariates can be checked, but potential outcomes are not fully observed. Researchers must use institutional knowledge and diagnostics to argue that no other discontinuous change occurs at the same cutoff.

9.3 Fuzzy RD requires a first stage

In fuzzy RD, the denominator $x^+ - x^-$ must be nonzero and meaningfully estimated. Weak discontinuities in treatment probability can lead to imprecise and unstable estimates, just as weak instruments do in IV.

9.4 Manipulation of the running variable

The paper focuses on identification under continuity assumptions and does not center the manipulation problem. If individuals can precisely manipulate Z around z_0 , the comparison of observations just above and below the threshold may no longer be credible. This concern motivates later density and sorting tests.

10 Bottom line

Hahn, Todd, and Van der Klaauw (2001) give RD its modern econometric foundation. The paper shows that the essential identifying comparison is the discontinuity in conditional means at the cutoff, that fuzzy RD is a local Wald ratio, that heterogeneous effects require careful interpretation, and that local linear regression is preferred because RD is a boundary problem. The paper's enduring contribution is the clean separation of identification, estimand interpretation, and estimation.